SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I S. Sasi Kumar (192324254) (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Sasi Kumar

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

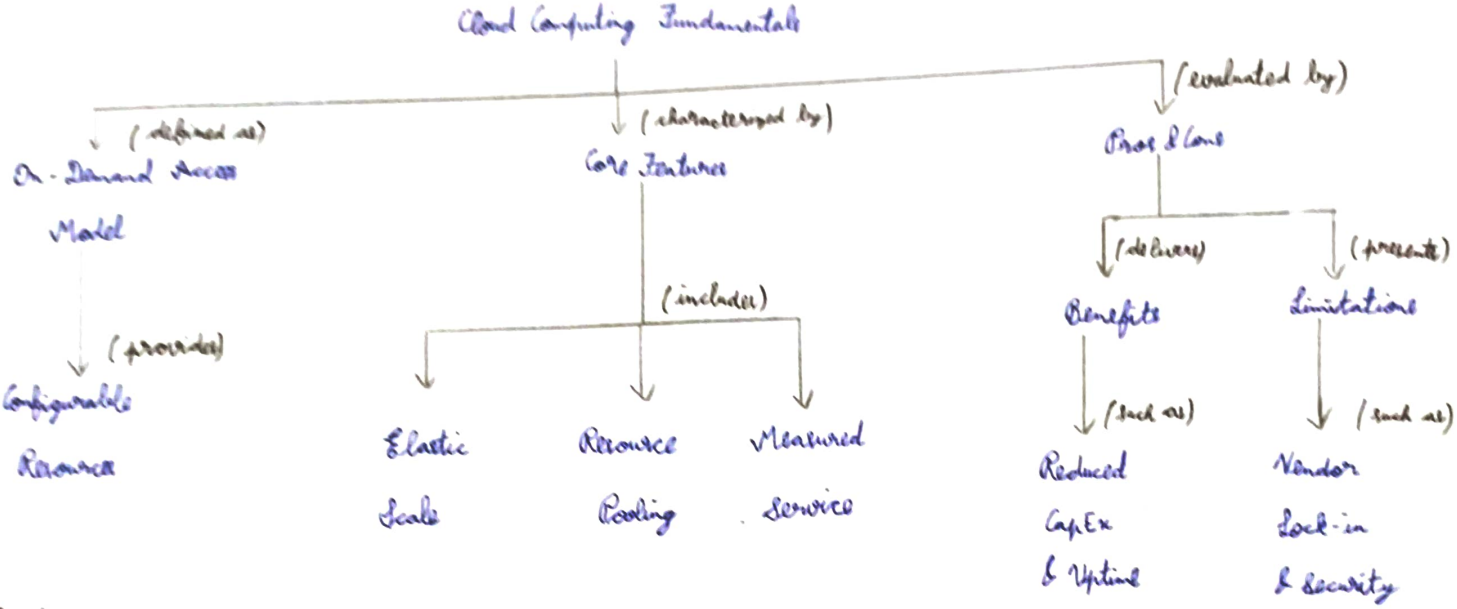
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

ASSESSMENT TOOL - 2

Concept Mapping

1. Cloud Computing Fundamentals



* **Definition:**

Model for on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications).

* **Core Characteristics:**

- On-demand access
- Resource Pooling
- Rapid elasticity
- Scalability.
- Measured service.
- Broad network access.

* **Benefits:**

- Reduced capital expenditure (CapEx to OpEx)
- High availability & disaster recovery.
- Global accessibility & speed to market.

* **Limitations:**

- Vendor lock-in
- Dependency on internet connectivity.
- Security, data privacy and compliance challenges.

Evolution of Cloud Computing



Hardware Evolution:

Mainframes → Commodity Servers → Multi-core & Blade Architectures → High density clusters.

Internet Software Evolution:

Monolithic Software → Web Services → Service-Oriented Architecture (SOA) → Microservices & APIs.

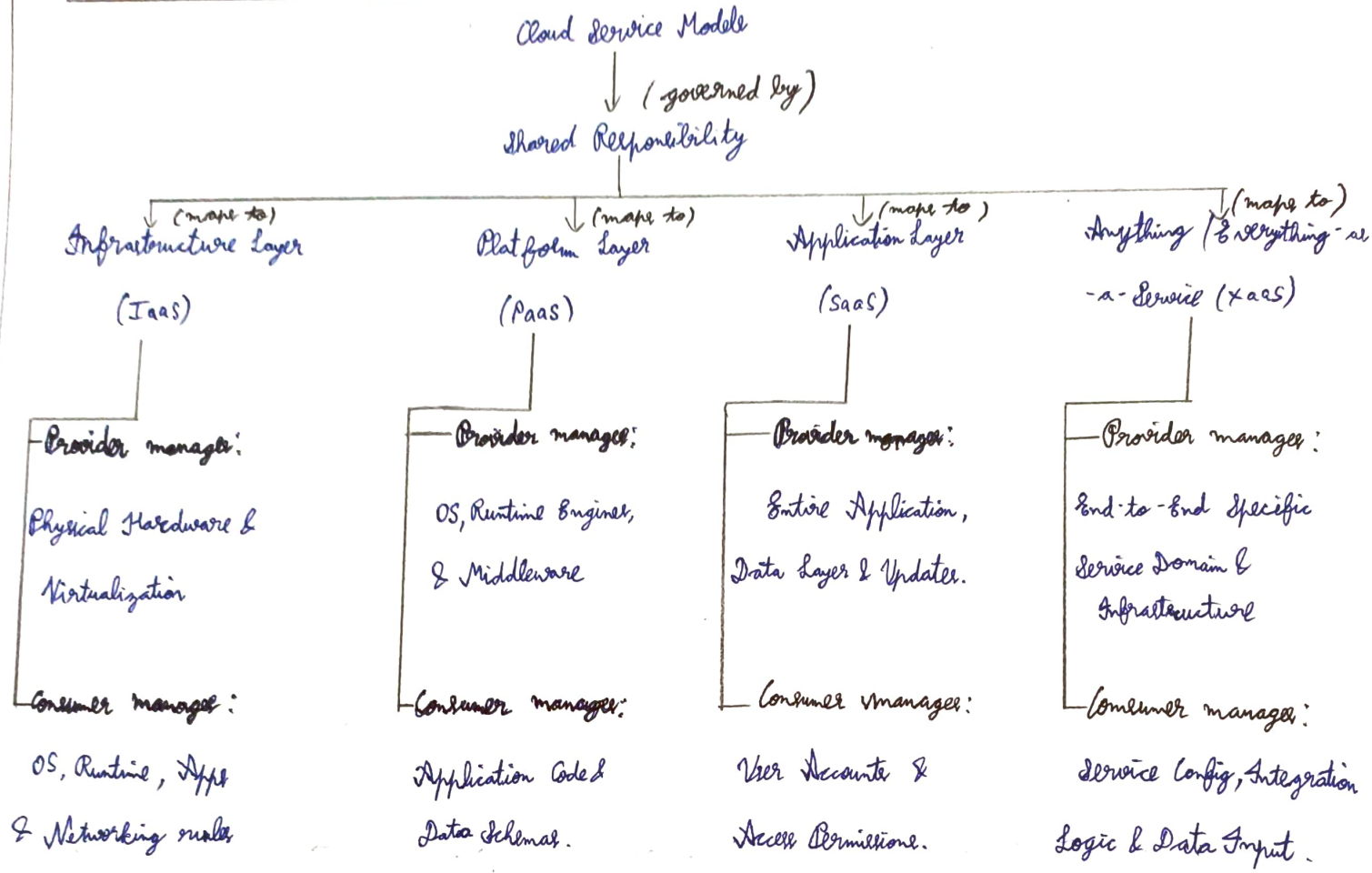
Distributed Systems:

Client-Server Architecture → Cluster Computing → Grid Computing → Utility Computing.

Service Delivery Models:

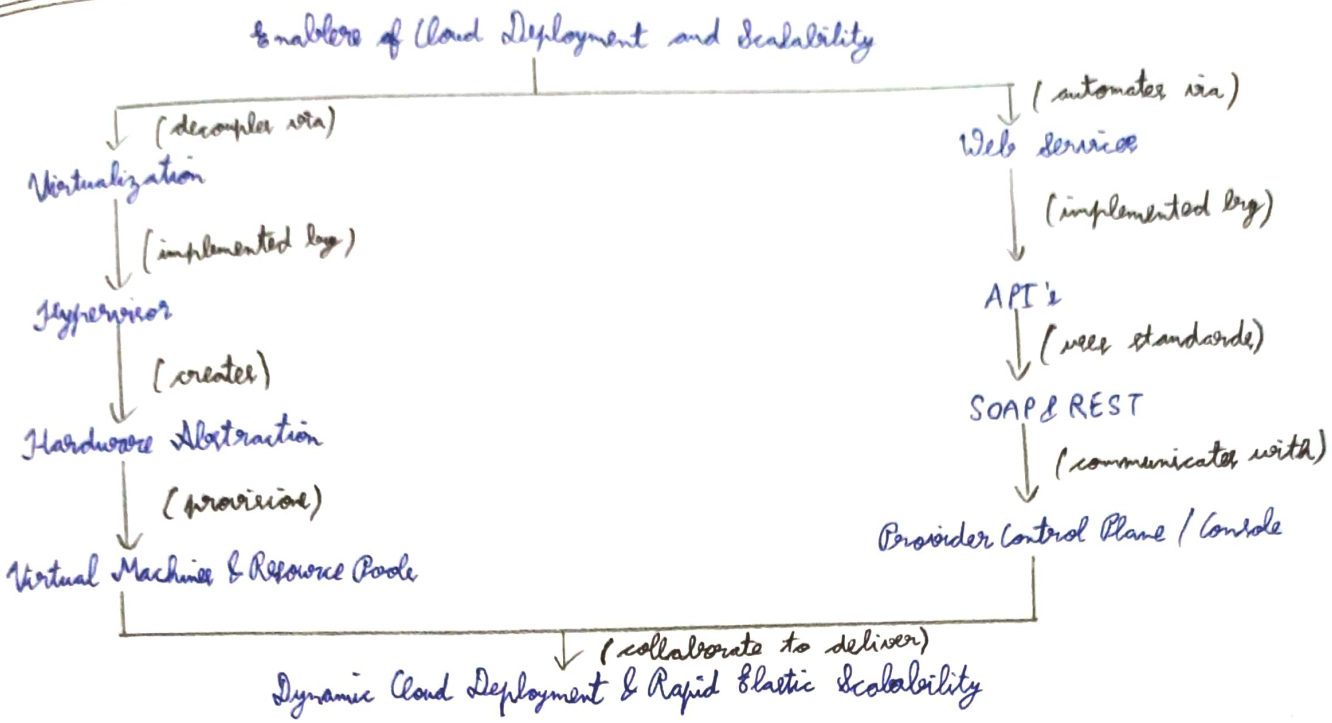
Time-sharing systems → Application Service Providers (ASPs) → Utility / Cloud Computing.

Service Models & Shared Responsibility.



* **Infrastructure as a Service (IaaS):**
  - **Provider Responsibilities:** Physical Data centers, server hardware & virtualization layer / hypervisor.
  - **User Responsibilities:** Operating system, middleware, runtime environment, application code, data, networking rules.
* **Platform as a Service (PaaS):**
  - **Provider Responsibilities:** Hardware, virtualization, OS, runtime engines, database engines, middleware.
  - **User Responsibilities:** Application development, application code, data schemas.
* **Software as a Service (SaaS)**
  - **Provider Responsibilities:** Full stack management (infrastructure, OS, runtime, application updates, backups).
  - **User Responsibilities:** User access controls, credential management, tenant data entry.
* **Anything / Everything as a Service (XaaS)**
  - Specialized cloud delivery covering functions such as FaaS (Serverless), DBaaS, and SECaaS.
  - Extends shared responsibility across micro-segmented cloud layers.

Virtualization and Web Services as Enablers



* **Virtualization:**

Uses a hypervisor to create an abstraction layer over physical hardware, generating virtual machines and pooled resources for multi-tenancy and workload isolation.

* **Web Services & API's:**

Exposes management endpoints using standard SOAP and REST protocols, enabling automated, programmatic control of computing resources.

* **Deployment & Scalability Impact:**

Virtualization provides the underlying resource elasticity, while web service API's automate rapid scaling and self-service deployment without manual hardware intervention.

13/10/26